MID-TERM EXAMINATION
(B.Tech AI&ML/ CSE-AI) 2nd semester
(May, 2023) OFF LINE mode

Subject Code: BAS 106	Subject: Environmental Sciences

Time : 1 ½Hours **Maximum Marks : 30**

Note: All parts of Q.1 are compulsory.

Q1		**(2.5*4)**	
	(a) What are various Environmental impacts of mineral extraction? Explain.		
	(b) What are the corrections needs to be considered while calculating gross calorific value (GCV) in Bomb Calorimeter		
	(c) The combustion process which tends to improve CO emissions can make the NOx emission problem worse and vice versa. Explain the statement. How will you control CO emissions?		
	(d) Explain the concept of 'Green starting materials', with a suitable example		
Q2	**(Attempt any Two Parts) UNIT-1**		**(5,5)**
	(a) Discuss the short term and long-term effects of ground water depletion?		
	(b) Write short note on (i) Under nutrition (ii) Malnutrition(iii) Bamboo Drip Irrigation System		
	(c) A Liquid fuel weighing 0.98 gm and containing 90.1% C, 8% H have the following results in bomb calorimeter experiment: Amount of water taken in calorimeter = 1,450 gm, water equivalent of calorimeter = 450 gm, rise in temperature = 1.8° C; Fuse wire correction= 10.0 cal; Acid correction=50.0 cal; Latent heat of condensation of steam as 580cal/g. Calculate gross andnet calorific value of fuel.		
Q3	**(Attempt any Two Parts) UNIT-2**		**(5,5)**
	(a) What is Acid Rain? Explain the process of formation of Acid Rain? How it affects the environment?		
	(b) List and explain the Tertiary treatment of waste water treatment? According to you which method under 3° treatment offers higher efficiency with low cost, Explain?		
	(c) Explain the concept of 'Prevention of waste'. Calculate the atom economy of conversion of Benzene to Maleic anhydride, giving a suitable chemical reaction		